PHISHGUARD

Phishing URL Risk Checker

Cybersecurity Bootcamp — Mini Project Report

Student Name: _____

Department / Institution: _____

Bootcamp: _____

Educational prototype for phishing awareness. Complete the student and institution fields before submission.

1. Abstract

PhishGuard is a command-line application written in Python that examines the text of a URL for selected warning signs associated with phishing. It applies transparent rules, calculates a simple risk score, and explains which checks contributed to the score. It is intended to teach basic cybersecurity awareness and is not a replacement for professional security tools.

2. Problem Statement

Phishing messages often use deceptive links to persuade people to disclose passwords, payment details, or other sensitive information. Beginners may not know which URL features deserve extra attention. A small educational tool can demonstrate common warning signs in an understandable way.

3. Objectives

- Demonstrate basic URL parsing in Python.
- Identify selected URL warning signs using simple rules.
- Display a risk category and human-readable reasons.
- Encourage safe link-handling habits without opening submitted URLs.

4. Scope

The prototype analyses the URL string entered by the user. It does not connect to the destination, download page content, check domain reputation, or use a machine-learning model. Its purpose is demonstration and learning.

5. Tools and Technologies

- Language: Python 3.8 or newer
- Libraries: Python standard library (`urllib.parse`, `ipaddress`, `re`, `sys`)
- Interface: command-line terminal
- Platform: any operating system with Python installed

6. Methodology

The user enters a URL. The program normalizes the input for parsing, extracts the hostname, and checks a list of indicators. Each detected indicator adds points to a simple score. The score is mapped to a category, and the program prints the reasons and a safety reminder. The tool does not visit the URL.

7. Indicators Used

- Non-HTTPS scheme: adds a caution point.
- '@' in the network location: may obscure the apparent destination.
- IP address used as hostname: may be suspicious in some contexts.
- Very long URL, many hyphens, or many subdomain levels.
- Encoded characters and selected words such as login, verify, or password.

These are imperfect signals: legitimate sites can trigger them and phishing sites can avoid them.

8. Risk Scoring

A score of 0–1 is labelled LOWER RISK SIGNALS, 2–4 is labelled CAUTION, and 5 or more is labelled HIGH RISK. The thresholds are illustrative and have not been validated against a labelled dataset. The labels are not proof of safety or maliciousness.

9. How to Run

1. Install Python 3.8 or newer.
2. Place `phishguard.py` in a folder.
3. Open a terminal in that folder.
4. Run `python phishguard.py`.
5. Enter a URL and review the score and reasons. Type `quit` to exit.

No third-party packages are required.

10. Testing Plan

Test with different inputs: a well-known HTTPS URL, an HTTP URL, a URL containing suspicious words, a long URL, an IP-address hostname, and blank input. Record the displayed risk category and the reasons. Testing should be done with harmless example URLs; do not visit suspicious links.

11. Expected Outcome

The program demonstrates how rule-based URL screening can flag potential warning signs and explain the result. It also shows the limitations of simple detection: false positives and false negatives are possible.

12. Limitations

• Rules are basic and can be bypassed.
• No external reputation or threat-intelligence checks are performed.
• The score is not statistically validated.
• The program cannot confirm whether a website is safe.
• It does not inspect email content, redirects, certificates, or website behaviour.

13. Future Enhancements

Possible future work includes a larger labelled dataset, carefully evaluated machine-learning features, a safe reputation API, a graphical interface, unit tests, and improved internationalized-domain handling. Any network feature should be designed with privacy and safe-use safeguards.

14. Conclusion

PhishGuard is a small, explainable Python project that introduces URL parsing, rule-based detection, scoring, and cybersecurity awareness. It provides a foundation for further study while clearly communicating that a heuristic result is not a security guarantee.

15. References

• Python documentation — `urllib.parse`: https://docs.python.org/3/library/urllib.parse.html

• Python documentation — `ipaddress`: https://docs.python.org/3/library/ipaddress.html
• CISA — Avoiding Social Engineering and Phishing Attacks: https://www.cisa.gov/secure-our-world/recognize-and-report-phishing